

SQL Assignment - 03

Course: Database Management Systems

Instructor: Ahmed Shahzad

Due Date: 1 PM, Sept 20th (Friday), 2025

Assignment Overview

This assignment focuses on applying intermediate SQL concepts to a realistic e-commerce database scenario. You will work with a comprehensive database schema that includes customers, products, orders, and related entities.

Database Schema

The e-commerce database consists of the following tables:

- **customers** - Customer information
- **categories** - Product categories with hierarchical structure
- **products** - Product catalog
- **orders** - Customer orders
- **order_items** - Individual items within orders
- **reviews** - Product reviews and ratings

Sample Data

Customers Table

customer_id	first_name	last_name	email	city	country	registration_date
1	John	Smith	john.smith@email.com	New York	USA	2023-01-15
2	Sarah	Johnson	sarah.j@email.com	London	UK	2023-02-20
3	Mike	Brown	mike.brown@email.com	Toronto	Canada	2023-03-10
4	Emily	Davis	emily.davis@email.com	Sydney	Australia	2023-01-25

5	David	Wilson	david.w@email.com	Berlin	Germany	2023-04-05
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Categories Table

category_id	category_name	parent_category_id
1	Electronics	NULL
2	Clothing	NULL
3	Books	NULL
4	Smartphones	1
5	Laptops	1
6	Men's Clothing	2
7	Women's Clothing	2

Products Table

product_id	product_name	category_id	price	stock_quantity	created_date
1	iPhone 14	4	999.99	50	2023-01-01
2	MacBook Pro	5	1999.99	25	2023-01-15
3	Cotton T-Shirt	6	29.99	100	2023-02-01
4	Programming Book	3	49.99	75	2023-01-20
5	Samsung Galaxy	4	899.99	40	2023-02-10

Orders Table

order_id	customer_id	order_date	total_amount	status
1	1	2023-02-15	1029.98	Delivered
2	2	2023-03-01	79.98	Shipped
3	1	2023-03-20	1999.99	Processing
4	3	2023-04-10	49.99	Delivered
5	4	2023-04-15	899.99	Pending

Order_Items Table

order_item_id	order_id	product_id	quantity	unit_price
1	1	1	1	999.99
2	1	3	1	29.99
3	2	3	2	29.99
4	2	4	1	49.99
5	3	2	1	1999.99

Reviews Table

review_id	product_id	customer_id	rating	review_date
1	1	1	5	2023-02-20
2	1	2	4	2023-03-01
3	2	1	5	2023-03-25
4	3	2	3	2023-03-05
5	4	3	4	2023-04-15

Questions

Part A: Basic Queries (30 points)

Question 1 (5 points)

Write a query to display all customers from the USA, ordered by their last name in ascending order.

Question 2 (5 points)

Find all products with a price greater than \$100, showing only the product name and price.

Question 3 (5 points)

List all orders placed in 2023, including the customer's first name, last name, and order date.

Question 4 (5 points)

Show the total number of products in each category, ordered by the count in descending order.

Question 5 (5 points)

Find all customers who registered in January 2023, displaying their full name and email.

Question 6 (5 points)

Display all products that are out of stock (`stock_quantity = 0`), showing the product name and category name.

Part B: Intermediate Queries (40 points)

Question 7 (8 points)

Write a query to show the top 5 customers by total order value, including their name, email, and total amount spent.

Question 8 (8 points)

Find all products that have never been ordered, displaying the product name and category name.

Question 9 (8 points)

Create a query that shows the monthly sales summary for 2023, including the month, number of orders, and total revenue.

Question 10 (8 points)

List all customers who have placed more than 2 orders, showing their name, email, and order count.

Question 11 (8 points)

Write a query to find the average order value by country, ordered from highest to lowest.

Part C: Advanced Queries (30 points)

Question 12 (10 points)

Using ROW_NUMBER(), create a query that ranks products within each category by price (highest to lowest). Show the product name, category name, price, and rank.

Question 13 (10 points)

Write a complex query that shows customers along with their most recent order details. Include customer name, most recent order date, order total, and the number of items in that order.

Question 14 (10 points)

Create a query that identifies products with above-average ratings (considering all reviews). Show the product name, average rating, number of reviews, and category name. Only include products with at least 3 reviews.

Bonus Questions (Extra Credit - 20 points)

Bonus Question 1 (10 points)

Write a query that shows the customer retention rate by month. For each month in 2023, show the percentage of customers who made a repeat purchase (had more than one order).

Bonus Question 2 (10 points)

Create a query that identifies potential high-value customers. Define high-value as customers whose total spending is in the top 20% of all customers. Show their details and suggest a marketing strategy.

Additional Notes

- If you encounter any issues with the schema, make reasonable assumptions and document them
- Focus on writing clean, readable, and efficient SQL code
- Test your queries with the provided sample data

Good luck with your assignment!
